

Exercise 4:

Reducing ODE:

$$y(1) = x$$

$$\frac{d(y(1))}{dt} = \frac{dx}{dt}$$

$$y(2) = \frac{d(y(1))}{dt}$$

$$\frac{d(y(2))}{dt} = \frac{d^2(y(1))}{dt^2}$$

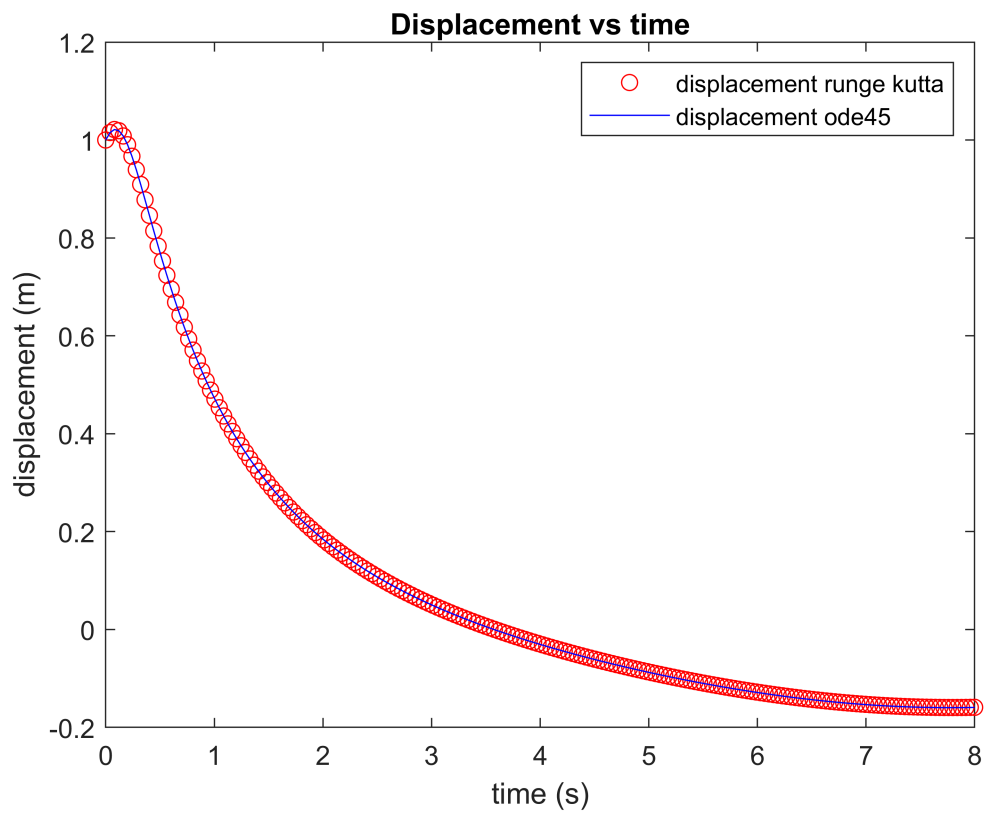
Then our system becomes:

$$m * \frac{dy(2)}{dt} + a * \text{sign}(y(2)) * (y(2))^2 + b * y(1)^3 = 0$$

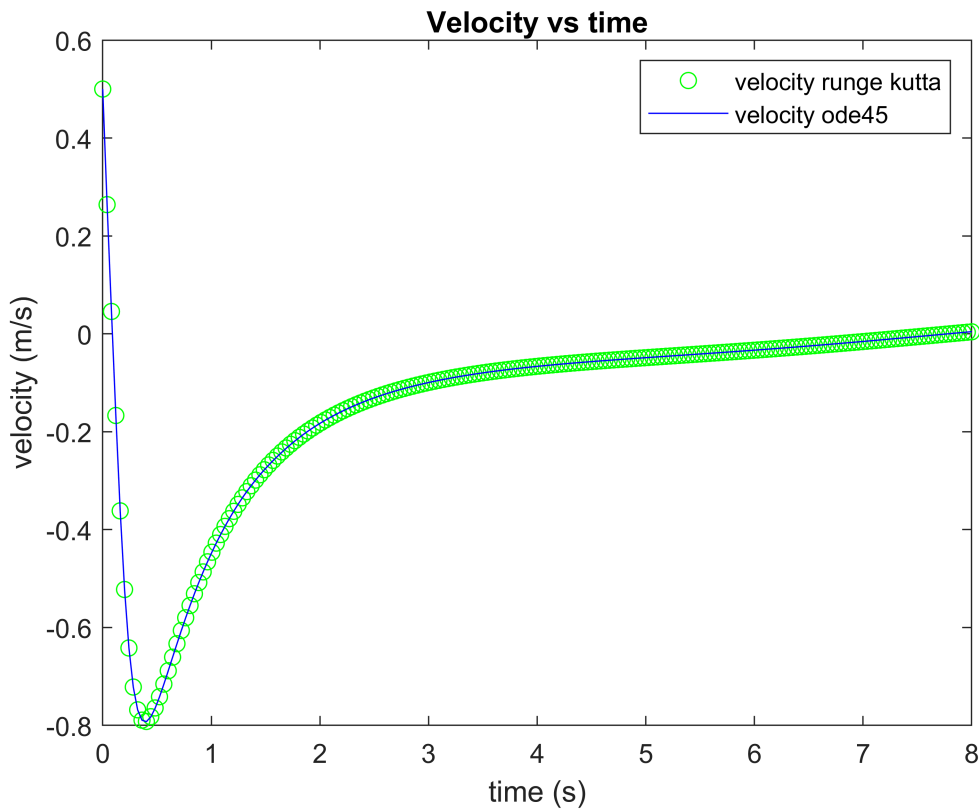
```
initialcond=[1; 0.5];
t=linspace(0,8,200);
func=@disp;
[result]=runge_kutta(func,t,initialcond);
displacement = result(1,:);
velocity = result(2,:);

[a b]=ode45(func,t,initialcond);
displacement_ode45=b(:,1);
velocity_ode45=b(:,2);

plot(t,displacement,'ro',t,displacement_ode45,'-b')
title('Displacement vs time');
legend('displacement runge kutta','displacement ode45');
xlabel('time (s)');
ylabel('displacement (m)');
```



```
plot(t,velocity,'go',t,velocity_ode45,'-b')  
title('Velocity vs time');  
legend('velocity runge kutta','velocity ode45');  
xlabel('time (s)');  
ylabel('velocity (m/s)');
```



```
function [dydt]=disp(t,y)
dydt=zeros(2,1);
m=1;a=5;b=5;
dydt(1)=y(2);
dydt(2)=(-1/m)*(a*sign(y(2))*y(2)^2+b*y(1)^3);
end
```

```
function [y] = runge_kutta(df,x,y0)
m = length(y0);
n = length(x);
% y is a matrix of size 'mxn'
y=zeros(m,n);
y(:,1)=y0;
for i=1:n-1
    h = x(i+1)-x(i);
    xi=x(i);
    yi=y(:,i);
    k1 = feval(df,xi,yi);
    k2 = feval(df,xi+0.5*h,yi+0.5*k1*h);
    k3 = feval(df,xi+0.5*h,yi+0.5*k2*h);
    k4 = feval(df,xi+h,yi+k3*h);
    y(:,i+1) = y(:,i) + h*(k1+2*k2+2*k3+k4)/6;
end
```

end